Final Grading Criteria: Real-Time IoT Data Pipeline

Oct 31, 2026 · @Rana Shaaban

The guidelines break the final marks into four areas: documentation, implementation, testing, and presentation. The mark for each area is not yet known, so the Marks column says TBD until the lecturer confirms the weights.

1. Mark Breakdown

Area	What Is Assessed	Evidence We Will Submit	Marks
Documentation	Project planning and management, literature review, requirements gathering, system analysis and design	The five planning files, Literature Review, Requirements Gathering, and System Analysis and Design, all uploaded to GitHub	TBD
Implementation	Source code and execution of the pipeline	Simulator, Flask API and Kafka producer, batch consumer, ETL to Snowflake, streaming alerts, and dashboard code	TBD
Testing	Validation of the data, alerts, and performance	Data validation report, alert test output, and KPI results	TBD
Presentation	Final presentation and reports	Slides, final project report, system architecture summary, and a live demo or dashboard screenshots	TBD
Total			TBD

2. Self-Check by Criterion

The team uses this list to check each area before submission. Owners follow the Task Assignment and Roles document.

Area	Criterion	How We Show It	Owner
Documentation	Planning files are complete and consistent	Proposal, Plan, Roles, Risks, and KPIs match each other	Rana
Documentation	Literature review and requirements are complete	Literature Review and Requirements Gathering with sources and sign-off	Rana
Documentation	Each component is documented	A README per component and the weekly documentation deliverables	Each owner
Documentation	All documents are on GitHub	Repository with a clear folder structure	Rana
Implementation	Data simulation works	Python generator script and sample data logs	Mostafa
Implementation	Ingestion works	Flask API and Kafka producer sending readings to the topic	Waleed
Implementation	Batch path works	Batch consumer, batched files, and ETL into Snowflake	Abdelrahman, Shahd
Implementation	Streaming alerts work	Streaming consumer with threshold rules and alert output	Mostafa
Implementation	Dashboard works	Trends, anomaly counts, and alert history on a live view	Rana
Testing	Loaded data is correct	Data validation report with completeness and accuracy results	Shahd
Testing	Alerts are correct	Test with injected anomalies and recorded results	Mostafa

Area	Criterion	How We Show It	Owner
Testing	Performance meets the KPIs	KPI results table for latency, uptime, and completeness	Team
Presentation	Final report is complete	Final project report and architecture summary	Rana compiles
Presentation	Presentation is ready	Slides and a rehearsed demo	Team

3. Submission Deadlines

These dates come from the Project Documentation Guidelines. The column that applies depends on the team's track.

Item	DEPI Industry	DEPI
Project Planning and Management	9 Oct 2026	31 Oct 2026
Literature Review	9 Oct 2026	31 Oct 2026
Requirements Gathering	9 Oct 2026	31 Oct 2026
System Analysis and Design	9 Oct 2026	6 Nov 2026
Implementation (Source Code and Execution)	16 Oct 2026	30 Nov 2026
Final Presentation, Testing and Reports	16 Oct 2026	4 Dec 2026

4. Open Questions for the Lecturer

- How many marks does each of the four areas carry?
- Are marks given to the team as a whole, to each member, or both?
- What rating scale is used in the Feedback and Evaluation form?
- Is a live demo required, or are dashboard screenshots enough?